MIL-F-20236D(SHIPS)
1 August 1963
SUPERSEDING
MIL-F-20236C(SHIPS)
7 October 1959

MILITARY SPECIFICATION

FITTINGS, TUBE AND PIPE, BUTT-WELDING, 300 P.S.I. AND 775°F. MAXIMUM

1. SCOPE

1.1 Scope.- This specification covers tube and pipe butt-welding fittings for:

(a) Steam services up to 300 pounds per square inch (p.s.i.) maximum working pressure and 775°F. maximum temperature.
(b) Oil and water services up to 400 p.s.i. maximum working pressure and 425°F. maximum temperature.
(c) Other services shown in MIL-STD-438 and MIL-STD-777.

1.2 Classification.- Butt-welding fittings shall be of the following types and classes as specified (see 6.2 and table II):

Type I - 90 degree and 45 degree elbows and return bends with tangents, tees and caps (see figure 1); concentric and eccentric reducers (see figure 2); branch outlets (see figure 3).
Type II - 90 degree and 45 degree elbows and return bends without tangents (see figure 4).
Class 1 - For tubing (150 p.s.i. maximum W.S.P.)
Class 2 - For tubing (300 p.s.i. maximum W.S.P.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY
MIL-Z-17871 - Zinc-Coating (Hot-Dip Galvanizing).

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy.
MIL-STD-438 - Schedule of Piping, Valves, Fittings and Associated Piping Components for Submarine Service.
MIL-STD-777 - Schedule of Piping, Valves, Fittings and Associated Piping Components for Surface Ships.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN STANDARDS ASSOCIATION (ASA)
B16.9 - Steel Butt-Welding Fittings
B16.25 - Butt-Welding Ends

FSC 4730

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(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York 16, New York.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 17, New York.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material.- The material shall be suitable for welding. Fittings shall be hot-formed and welded; hot-forged; cold-drawn; pressed; or machined. Fittings shall be of steel. Unless otherwise specified in the contract or order, the material shall be carbon steel.

3.2 Chemical composition.-

3.2.1 Wrought-steel fittings.- Wrought-steel fittings shall be of the chemical composition specified in table I.

Table I - Chemical composition for wrought-steel fittings.

Carbon (max.)	Manganese	Phosphorus (max.)	Sulphur (max.)	Other elements each (not including iron) (max.)
Percent	Percent	Percent	Percent	Percent
0.31	0.29 - 1.06	0.048	0.050	0.25

3.3 Sizes.- The dimensions and tolerances of the fittings shall be strictly as specified herein.

3.4 Fittings fabricated by welding.- Fittings shall be welded in accordance with MIL-STD-278. Fittings made from plates, sheets or strips shall not have more than two longitudinal welded seams. No fittings shall be made with any circumferential welded seam. (This does not refer to partially circumferential seams incident to fabrication with branch outlets, (see figure 3).

3.5 Fittings which are cold-drawn, or fabricated by hot-forming or by welding, shall be stress-relieved after fabrication in accordance with MIL-STD-278. The stress-relieving operation shall be performed before final machining.

3.6 Fittings shall be black unless zinc-coated (galvanized) fittings are specified (see 6.2). Zinc coating shall be done by the hot-dip process, in accordance with MIL-Z-17871, after fittings are fabricated and cleaned. Black fittings shall be protected with a rust-preventive compound approved by the bureau or agency concerned.

3.7 Fittings shall be cut to the proper angle, and shall have full circular cross-sectional area throughout. The ends of the fittings shall be prepared in accordance with ASA B16.9 and B16.25.

3.8 Manufacturers shall certify that all fittings offered for delivery are capable of withstanding without leakage, a test pressure of 500 p.s.i. for class 1 and 750 p.s.i. for class 2.

3.9 Flattening shall be as specified in 4.4.2.

3.10 Dimensions.- The nominal dimensions of all fittings shall be as shown in ASA B16.9, except as specified in 3.10.1. The tolerances on fittings and dimensions shall be in accordance with ASA B16.9. The wall thickness shall be as shown in table II.

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Table II - Dimensions for elbows, tees, return bends and caps (see figures 1 and 4).

Size	O.d.	Thickness "T"	
		Class 1	Class 2
		150 p.s.i.	300 p.s.i.
	Inches	Inch	Inch
1-1/4	1.660	0.120	0.120
1-1/2	1.900	.120	.120
2	2.375	.120	.120
2-1/2	2.875	.120	.120
3	3.500	.120	.120
3-1/2	4.000	.120	.134
4	4.500	.120	.134
5	5.563	.165	.165
6	6.625	.220	.220
1/ 7	7.625	.238	.238
8	8.625	.238	.238
1/ 9	9.625	.238	.259
10	10.750	.259	.259
1/ 11	11.750	.203	.380
12	12.750	.284	.425
--------	14	.340	.454
--------	1/ 15	.364	.482
--------	16	.380	.500

1/ For replacement only.

3.10.1 Unless otherwise specified (see 6.2), 90 degree and 45 degree elbows and return bends shall be furnished with tangents. The length of the tangents shall be as follows:

(a) For fittings 3-1/2 inches and below not less than 3/8 inch minus 0, for fittings above 3-1/2 inches not less than 1/2 inch minus 0.

(b) The length of the tangent shall be included in the center-to-end dimension. The center-to-end dimension shall be as shown in ASA B16.9.

3.10.2 Branch outlets (see figure 3).- Branch outlets forged, welding and socket ends, shall have the welding end for attachment to the branch machined to the dimensions shown in table III. The bore of the fittings shall be machined to the dimensions shown on figure 3 and in table III, within a tolerance of plus or minus 0.015 inch. The maximum size outlet shall be two sizes smaller than the pipe to which it is welded.

3.11 Marking.- Each fitting shall be legibly and permanently marked by means of electro-etch or electric pencil. The use of low stress stamping is permitted provided that the depth of the impression does not exceed 0.010 inch and the minimum wall thickness is maintained.

3.12 Workmanship.- Fittings shall be smooth, true to form, uniform in texture, sound and free from folds, piping, grooving, indentations, buckles, segregation, cracks, and other defects which might affect serviceability. Fittings shall be thoroughly cleaned both inside and outside, and all fins and roughness shall be removed. Workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Table III - Dimensions of forged welding outlets (see figure 3).

Main line size (I.p.s.)	R	Main line size (I.p.s.)	R
Inches	Inches	Inches	Inches
1-1/2	0.95	6	3- 5/16
2	1- 3/16	7	3-13/16
2-1/2	1- 7/16	8	4- 5/16
3	1- 3/4	9	4-13/16
3-1/2	2	10	5- 3/8
4	2- 1/2	11	5- 7/8
5	2-25/32	12	6- 3/8

Nominal size of outlet (I.p.s.)	A	B	C	D	I. D.		E
					Socket 1/	Butt 2/	
1/2	7/8	1-13/32	29/32	15/32	0.855	----	1-11/64
3/4	7/8	1-23/32	1- 5/32	1/2	1.065	----	1-25/64
1	1- 3/16	2- 1/8	1- 7/16	17/32	1.330	----	1-43/64
1-1/4	1- 5/16	2- 9/16	1- 3/4	11/16	1.675	----	2- 3/64
1-1/2	1- 5/16	2-27/32	2	3/4	1.915	----	2- 5/16
2	1- 1/2	3- 5/16	2- 9/16	-----	2.000	2.051	-------
2-1/2	1- 5/8	4- 5/64	3	-----	----	2.454	-------
3	1- 3/4	4-51/64	3-11/16	-----	----	3.053	-------
3-1/2	1- 7/8	5-15/32	4- 7/16	-----	----	3.533	-------
4	1-31/32	5-31/32	4- 3/4	-----	----	4.011	-------
5	2- 1/4	7- 5/16	6- 1/8	-----	----	5.032	-------
6	2- 7/8	8- 5/8	7- 1/4	-----	----	6.040	-------
8	2- 3/4	10-11/32	8-11/16	-----	----	7.956	-------

1/ Tolerance for socket: plus 0.010 inch, minus 0.000 inch.
2/ Tolerance for butt: plus 0.000 inch, minus 0.010 inch.

4.2 Sampling for quality conformance inspection.-

4.2.1 Lot.- For purposes of sampling, a lot shall consist of all fittings of the same type, class, size, form and heat.

4.2.2 Sampling for examination.- Sample fittings shall be selected from each lot in accordance with table IV and MIL-STD-105. The acceptable quality level equals 1.5 percent defective. Examination shall be conducted as specified in 4.3.

Table IV - Sampling for examination.

Lot size	Sample size (single sampling)	Acceptance number (defectives)	Rejection number (defectives)
180 and under	25	0	1
181 to 300	35	1	2
301 to 500	50	2	3
501 to 800	75	3	4
801 to 1,300	110	4	5
1,301 to 3,200	150	5	6
3,201 to 8,000	225	8	9
8,001 to 22,000	300	10	11
22,001 to 110,000	450	14	15
110,001 to 550,000	750	20	21
550,001 and over	1,500	35	36

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4.2.3 Sampling for chemical analysis.- Drillings shall be taken from two sample fittings from each lot for the chemical analysis specified in 4.4.1. Drillings may be taken from fittings subjected to flattening test. If facilities are not available at the place of manufacture analysis shall be made at a laboratory satisfactory to the Bureau of Ships.

4.2.4 Sampling for flattening test.- Two sample fittings shall be selected from each lot for the test specified in 4.4.2.

4.3 Examination.- Samples selected in accordance with 4.2.2 shall be visually examined and checked for concentricity and for conformity with the respective dimensions and tolerances in accordance with the requirements of this specification not involving tests.

4.4 Tests.-

4.4.1 Chemical analysis.- Sample drillings selected in accordance with 4.2.3 shall be analyzed to determine conformance with the composition specified in 3.2.1. If any sample fails to conform to the requirement specified, the lot represented by the sample shall be rejected.

4.4.2 Flattening tests.- Sample fittings, selected in accordance with 4.2.4, shall be flattened cold under pressure applied slowly between parallel surfaces until the distance between the surfaces is not over 1/3 the original fitting diameter. No cracks or openings shall be permitted under this test. If any sample fails to conform to this test, the lot represented by the sample shall be rejected.

4.5 Radiographic inspection.- When specified (see 6.2), welded joints in fittings shall be 100 percent radiographically inspected in accordance with MIL-STD-278.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation.-

5.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation, packaging, packing and marking.- Fittings shall be preserved and packaged by level A or C; packed and marked by level A, B or C as specified, in accordance with MIL-V-3.)

6. NOTES

6.1 Intended use.- Steel fittings are intended to be welded to seamless or welded steel pipe or tubing made to the following specifications:

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SPECIFICATIONS

FEDERAL
WW-P-404 - Pipe, Steel (Seamless and Welded, Black and Zinc-Coated (Galvanized)).

MILITARY
MIL-T-20157 - Tube and Pipe, Carbon Steel, Seamless (Type A and B Only).
MIL-T-20162 - Tubing, Steel, Welded (Including Resistance-Welded), for Oil, Steam or Water.
MIL-T-20169 - Tubing, Steel, Resistance-Welded, for (Oil, Steam or Water) (Types A, B and C Only).

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and class required (see 1.2).
(c) Whether zinc-coated (galvanized) fittings are required (see 3.6).
(d) Whether 90 and 45 degree elbows and return bends should be furnished with tangents (all Navy orders to provide tangents) (see 3.10.1).
(e) Whether radiographic inspection is required (see 4.5).
(f) Preservation, packaging, packing or marking requirements other than those required by 5.1 (see 5.2).
(g) Fittings should be ordered by the piece.
(h) Where fittings (other than outlets) are required of a thickness different from the respective thickness specified herein, the thickness and the outside diameter should be specified.
(i) Reduction in the run or outlet of a tee will be made using a reducer (figure 3) of the required size with a straight tee (figure 1).

Preparing Activity:
Navy - Ships
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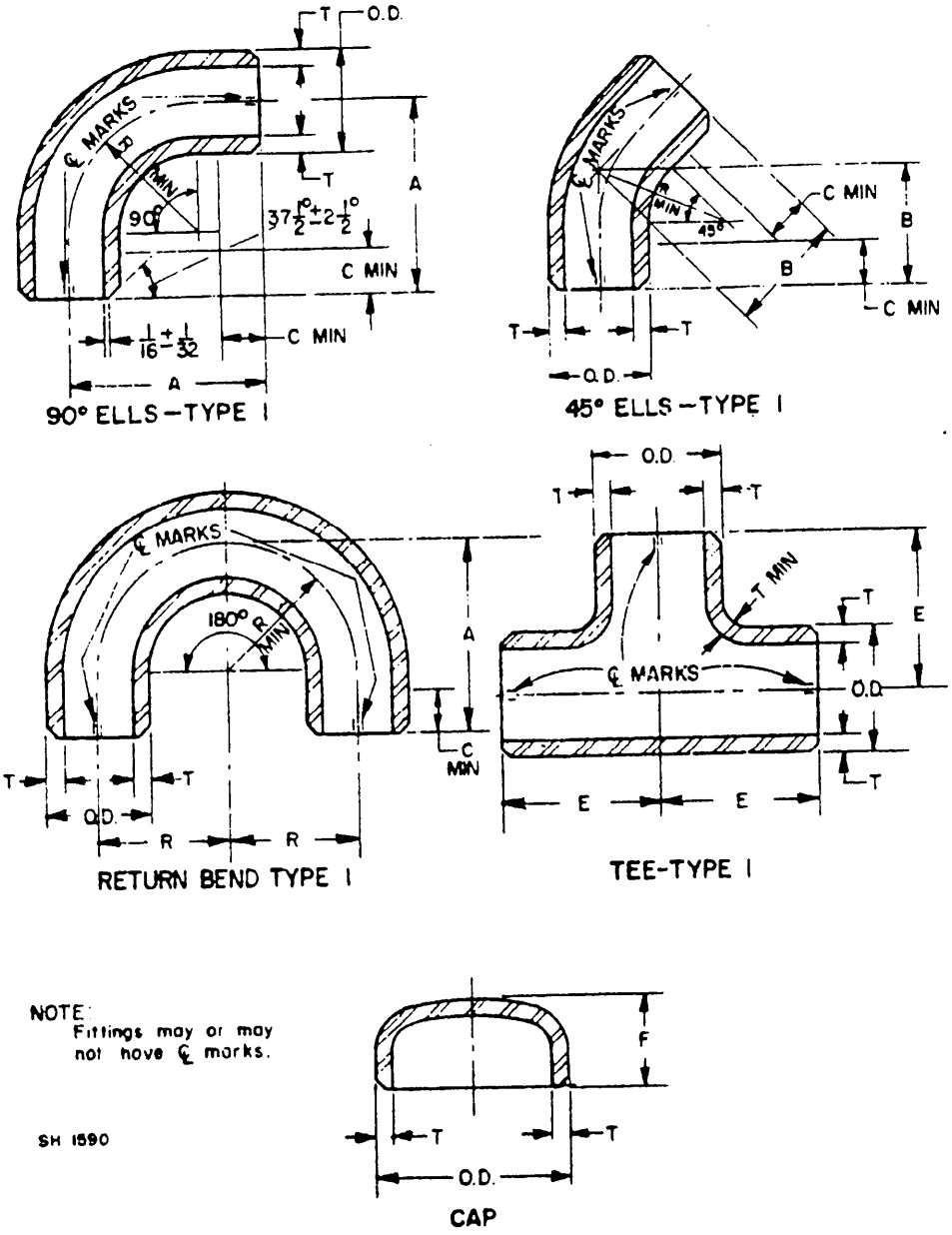
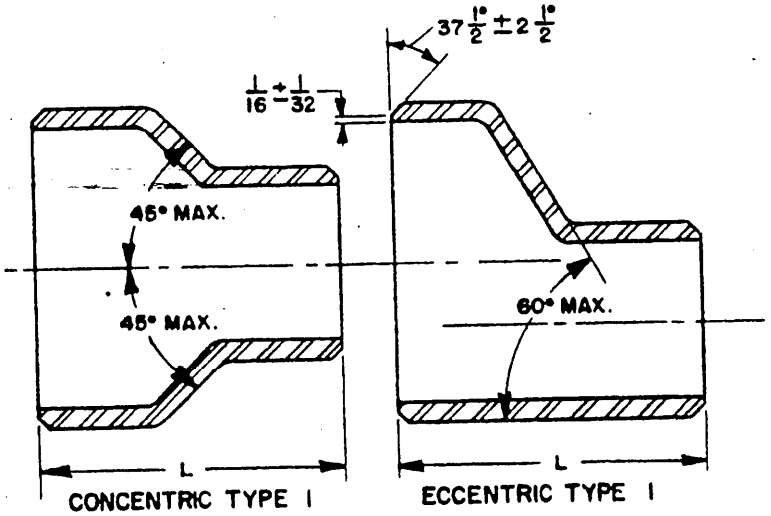


Figure 1 - Elbows, return bends with tangents, tee, and caps (see table II).

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NOMINAL PIPE SIZE	REDUCERS LENGTH "L" SIZE OF REDUCED END																
	1 1/4	1 1/2	2	2 1/2	3	3 1/2	4	5	6	7	8	9	10	11	12	14	15
1 1/2	2 1/2																
2	3	3															
2 1/2	3 1/2	3 1/2	3 1/2														
3	3 1/2	3 1/2	3 1/2	3 1/2													
3 1/2	4	4	4	4	4												
4		4	4	4	4	4											
5			5	5	5	5	5										
6				5 1/2	5 1/2	5 1/2	5 1/2	5 1/2									
7					5 1/2	5 1/2	5 1/2	5 1/2	5 1/2								
8					6	6	6	6	6								
9						6	6	6	6	6	6						
10							7	7	7	7	7	7					
11								7	7	7	7	7	7				
12									8	8	8	8	8	8			
14													13	13	13		
15													13	13	13	13	
16													14	14	14	14	14

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NOTE: SIZES 7, 9, 11, AND 15 ARE NOT COMMERCIAL.

Figure 2 - Concentric and eccentric reducers.

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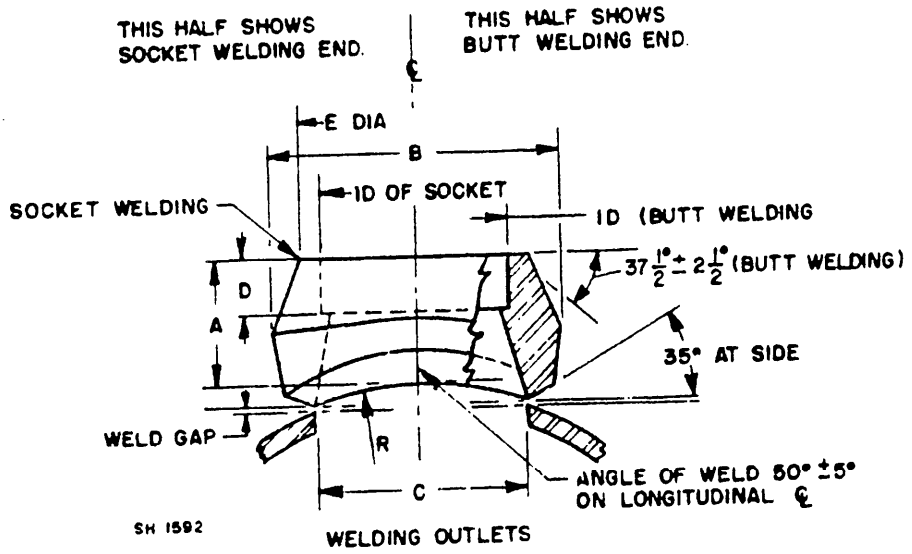


Figure 3 - Forged welding branch outlets (see table III).

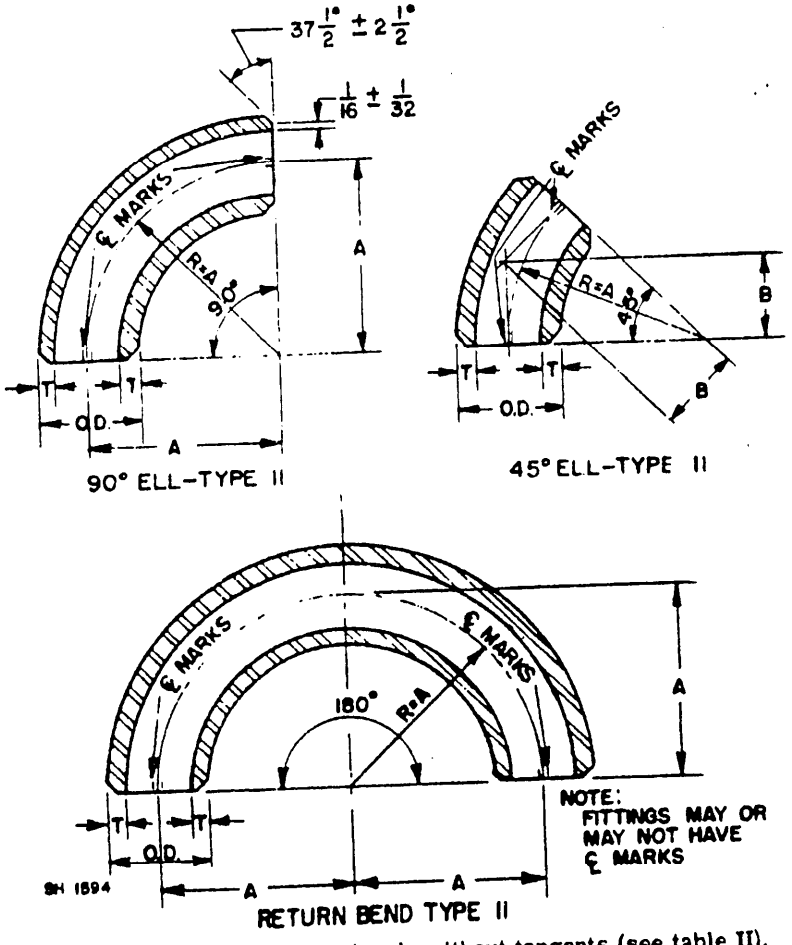


Figure 4 - Elbows and return bends without tangents (see table II).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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